CMSC 2621
Lab Exercise 5

Due: See the assignment folder for the due date.

Overview:

In this exercise, you will manage the operations of a busy space station using a Queue for handling the arrival of space shuttles and a Double-Ended Queue (Deque) for managing supplies and emergency priorities. By the end of this exercise, you will have a solid understanding of how queues and deques work in C++.

Requirements:

1. Download the SpaceStation.cpp file that's linked in the D2L assignment. Follow the instructions in the comments and update only the locations in the code that have a "CHANGE_HERE"
2. Submit the source code and a document of a screenshot in a zip file to the D2L assignment. (Please see the instructions in the end of this document)

Submission Requirements:

You'll be submitting a zip file that contains the files below:

1. The source code file
2. A Word document containing screenshots of your source code and the full results of executing your program.

After creating a zip file containing all the required files, submit the zip file to the assignment folder on D2L.

Screenshot Requirements:

1. The screenshot(s) need to be clear enough to read the source code.
2. If the running results are too long to be captured, use multiple screenshots.
3. The running result(s) need to be captured in the screenshot(s) completely. If the running results are not complete, i.e. just partial results are captured in the screenshot(s), you may also receive a points deduction, depending on how complete the results in the screenshots are.

Assignment Grading:

1. If the program fails to be compiled using a compiler that supports C++ 20, a maximum grade of 40% will be given.
2. If any requirements are not met, for each unfulfillment, a corresponding penalty will be given.
3. Completeness of the lab assignment will also be considered.

Example Output:

```
Shuttle Discovery added to docking queue.
Shuttle Atlantis added to docking queue.
Shuttle Endeavour added to docking queue.
Shuttle Challenger added to docking queue.
Shuttle Enterprise added to docking queue.
Queue is full! Can't add shuttle Columbia.
Docking shuttle: Discovery
Docking shuttle: Atlantis
Added Oxygen Tanks to the rear of the deque.
Added Food Supplies to the rear of the deque.
Added Medical Supplies to the front of the deque.
Using: Medical Supplies
Using: Food Supplies
Added Water Supplies to the rear of the deque.
Added Spare Parts to the rear of the deque.
Added Extra Fuel to the rear of the deque.
Using: Oxygen Tanks
Using: Extra Fuel
Using: Spare Parts
Using: Water Supplies
Deque is empty! Can't remove item from the rear.

Deque is empty! Can't remove item from the front.
```